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Review

Aspects of arc-flash protection and prediction

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ABSTRACT

This paper provides a technological review of arc-flash protection of air insulated switchgear. It covers the whole range starting from switchgear design aspects until ultra-fast arc elimination. Special attention is paid to proactive technologies enabling preemptive detection of slowly developing faults. Various arc faults indicating phenomena are examined, and several sensor technologies for online monitoring are evaluated. Because preventive or predictive measures cannot totally eliminate the risk of arc faults, reactive protection by fast operating protection is justified. Two major reactive protection approaches are discussed: the current-limiting approach and the arcing time based approach. The benefits of protection based on simultaneous detection of light and overcurrent are explained. Finally the paper discusses arc elimination technologies and evaluates the concerns related to short-circuit devices.

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1. Introduction

Internal high power arc fault in switchgear is a reasonably rare incident, but can have very serious consequences unless the arc is rapidly extinguished. An arc fault causes hazardous impacts: thermal impact, pressure wave, flying particles, electrical and toxic impact. In addition to the hazard to personnel, substantial economic losses due to equipment damage and long system interruptions are common consequences.

There are a number of options to limit the risks related to arc faults. Some of the approaches emphasize arc prevention while others focus on technologies for arc mitigation. This paper gives an extensive overview of the whole field. Emerging technologies in arc fault prevention and very fast reacting protection systems are discussed more in detail.

2. A holistic view of existing arc-flash protection technology

Various approaches aiming at prevention or mitigation of arc flash incidents have been presented. There are significant differences in the methodology, based on traditions and cultural difference. Standardization is only partially a reason for different practices. For example in Europe, in spite of the influence of IEC standards, there is no universal practice to deal with arc flash issues.

Arc-flash protection methods can be categorized into many ways. Examples of categorization are division into passive and active methods, and division into proactive and reactive methods. Passive methods do not have any active component. Active methods include measurements and either reactive response (after arc-flash has been detected) or proactive response (pre-arc action when indication of a developing fault has been detected).

Approaches aiming at prevention of arc-flash, like design of switchgear, education of personnel, and maintenance practices, can be categorized into pre-ignition methods and passive methods. Fig. 1 presents the categorization and a holistic view of arc-flash protection.

In the following sections, the elements of the holistic view are examined in more detail. Special attention is paid to emerging technologies of arc prediction, i.e. online monitoring in order to detect developing fault before they escalate into high power arc faults.

However, there are faults which cannot be predicted by online monitoring, e.g. faults caused by direct human interaction or by animals, or faults caused by incorrect operation of switching devices. Therefore, reactive arc-flash protection, operating after arc ignition, is often considered justified. Along with online monitoring technologies, efficient arc mitigation technologies are reported in detail.

3. Arc-flash prevention

3.1. Education

Either direct or indirect human action is a common cause of arc faults. Education of workers, safety culture and rules regarding operation in an environment with possible arc-flash hazard should be commonplace. Incident energy calculations and arc labeling of switchgear are a good example of safety culture.

Education of personnel is justified for safety and economic reasons. Arcing faults cause direct damage costs and costs due to process interruption, and when humans are involved, high medical or legal costs are possible.

3.2. Design

Prevention of internal arcs in switchgear starts naturally in design. Requirements have been set in IEC and IEEE standards [1–4]. The IEC standard [1] presents a list of locations where internal arc faults are most likely to occur in metal-enclosed switchgear and controlgear:

- Connection compartments
- Disconnectors, switches, grounding switches
- Bolted connections and contacts
- Instrument transformers
- Circuit breakers

The standard also lists possible causes of internal arc faults and examples of measures to decrease the probability of faults. These measures include both technical solutions and personnel related recommendations. Some additional design options are evaluated below.

3.3. High resistance grounding

High resistance grounding, limiting phase-to-ground fault currents, can be seen as an arc-flash preventive technique. It reduces drastically the dissipated energy, and sustained arcing faults in low voltage (LV) systems. This is why proponents of this system have proposed that it should become a standard of the industry [5].

However, high-resistance grounding is only effective in ground faults and requires that the first ground fault can be cleared before the second ground fault causes phase-to-phase fault [6]. Although majority of arc faults start as phase-to-ground faults, other protection means are necessary when high resistance grounding is applied. Additionally, high-resistance neutral grounding seems to have very limited window of application on medium voltage (MV) systems for several reasons explained in [7].

3.4. Insulated vs. bare busbar

Insulated bus appears to be superior to bare bus, but there are different opinions. First, it is obvious that insulation provides means to reduce the probability of arc faults caused e.g. by falling objects or by vermin. Another advantage is that insulation prevents single-phase faults from escalating to high power multi-phase faults [8].

The third possible advantage is that the arc may travel to insulated area and become self-extinguishing [9]. This is, however, controversial. In tests and real arc faults reported in [10], the arc

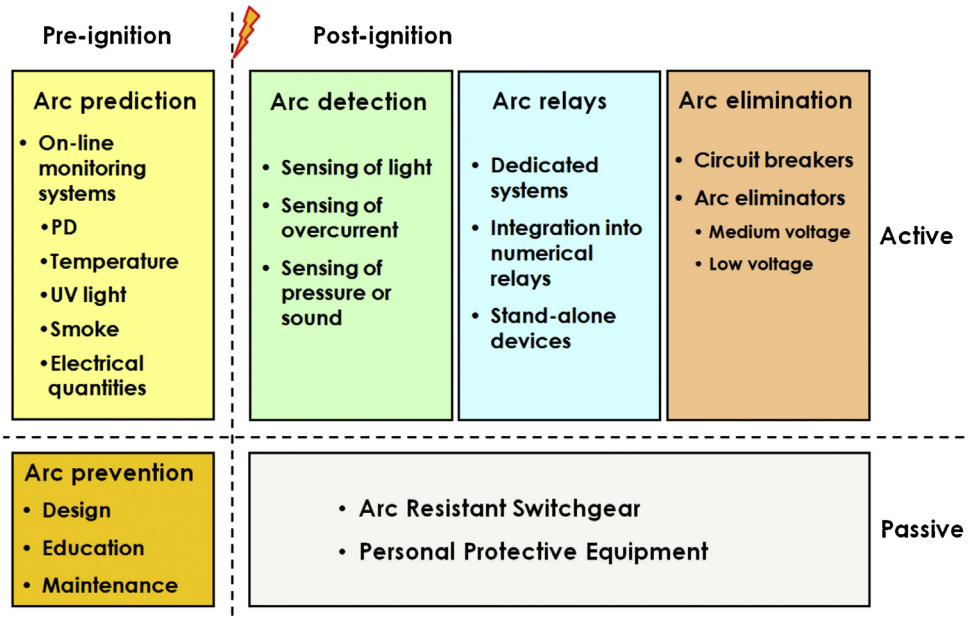


Fig. 1. The categorization and the holistic view of arc-flash protection.

moved slowly and burned through the insulation, causing more damage. The insulation does not necessarily extinguish the arc.

It has also been noticed in partially insulated switchgear tests that the movement of the arc, which is fast with bare bus, stops at the starting point of the insulation. Somewhat stable arc at the insulating barrier point causes melting and vaporization of metal, leading to more significant damage compared to moving arc.

Fig. 2 illustrates the impact of a stationary arc in the point where bus insulation starts. The figure presents LV busbars after an arc-flash test with fast protection. It must be noted that in this case the buses are only partially insulated. The insulation does not cover the joint area. Although the duration of the arc in this test was only some tens of milliseconds, the arc melted some of the busbar metal at the insulating barrier point.

This observation is in line with the one made in [10]: if plasma is allowed to concentrate, the rate of damage will accelerate. Also according to low voltage testing reported in [11], an insulating barrier prevents arc motion, and produces higher arcing currents and higher incident energy. The barrier effect was increased at the higher test voltages.

3.5. Maintenance

The probability of arc faults can be lowered by appropriate preventive maintenance of equipment. Visual inspection, thermal imaging, partial discharge testing, and time-based testing of protection devices are examples of preventive maintenance actions. Preventive maintenance is effective against aging, corrosion, pollution or vibration related faults. Condition monitoring of equipment, often by on-line measurements and communication to upper level information systems, is a very interesting option to prevent arc faults.

4. Proactive protection: arc-flash prediction technologies

4.1. Arc-flash prediction

Loose connections and degradation of insulation are common causes of arc faults. These fault types develop gradually, and it may be possible to analyze pre-arc conditions and find early signs of developing faults. This can be called arc fault prediction. Existing technologies enable on-line monitoring of electrical apparatus. A combination of different monitored physical phenomena adds reliability to detection of developing faults [12].

In the following, first an overview is given of potential arc fault prediction methods which are based on common current and voltage measurements, then other phenomena and sensor technologies are discussed.

4.2. Detection of developing faults by analysis of phase currents

Since current is practically always measured and analyzed anyway for protection purposes, it would be very convenient to use normal phase current measurements to find developing faults. Low current arcs, preceding actual high current and high power arcs in low voltage systems (e.g. in loose connections), have been studied.

In [13], the harmonic spectrum of the phase currents in a low voltage system was analyzed. The focus was on 3rd, 5th and 7th harmonic. The paper concludes that it is generally possible to design an operable preventive protection.

Low voltage systems and harmonics were also discussed in [14]. The paper examines the characteristics of low current arc faults which can precede high power arc faults. A detailed harmonic analysis is presented for higher frequencies and the frequencies in between the harmonics. In lower harmonics area the third harmonics could be an indicator of lower power arc faults. In higher frequency area up to 2 kHz, the broadband signal of the arc can be a strong indicator. In the range of 2–5 kHz, the re-ignition of the parallel arc causes slight elevation in the current spectrum. However, because some loads generate harmonics and increased higher frequency components, this detection method cannot be the only recognition criterion of developing arc faults.

4.3. Detection based on zero sequence voltage or current differential

In MV systems with ungrounded or compensated neutral, changes in zero sequence voltage can give an indication of a developing ground fault. This indication is commonly used by electric utilities in some countries for early detection of high-resistance ground faults. However, for detection of developing

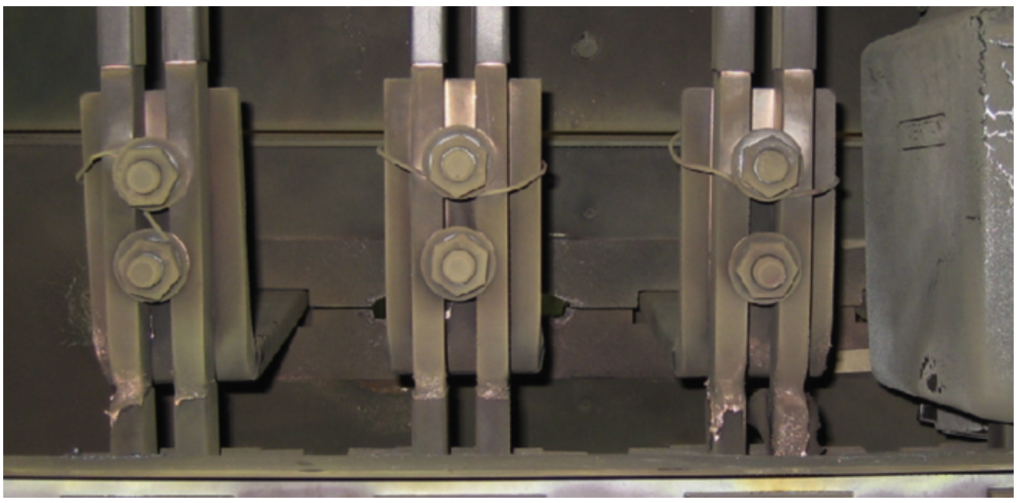


Fig. 2. Partial melting of LV busbar after short-time stationary arcing at the insulation barrier. (U = 726 V, I_k = 77 kA, t_{arc} = 65 ms.)

faults in switchgear, monitoring of zero-sequence voltage should be used in combination with other indication methods because it does not reveal the location of the fault.

Cable termination is a typical location of arc faults. Principle of current differential protection may be applied for detection of developing faults in cable terminations [15]. In practice this means comparison of the sum of individual phase current measurements and the measurement of the core balance current transformer. Fig. 3 illustrates this principle.

4.4. Other phenomena indicating developing arc fault

In addition to commonly measured electrical quantities, there are various physical quantities that can potentially indicate an arc development. The utilization of these phenomena often requires sensor installations in the switchgear. Potential physical phenomena have been listed and shortly described below, along with their detection methods. Detailed evaluation of them has been reported in [16,17].

(1) *Electromagnetic emissions*: Load current in loose connection can cause micro sparks and ionization of the air. Partial discharges (PD) and corona also cause ionization. Ref. [18] gives an extensive review of PD measurement techniques in gas-insulated switchgear while this paper focuses on air insulated switchgear.

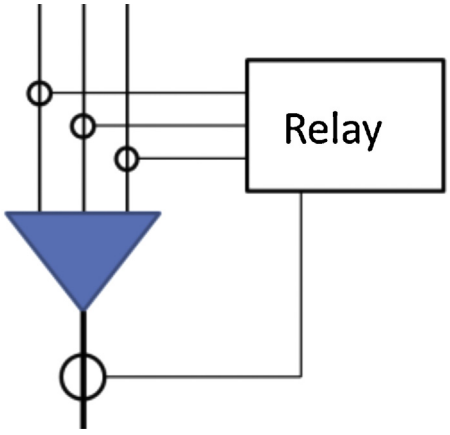


Fig. 3. I_{diff} monitoring of cable termination.

(2) *Acoustic (ultrasonic) emissions*: PDs also cause mechanical vibrations in electrical equipment. An acoustic signal in ultrasonic range of spectrum is emitted as a result of such vibration.
(3) *Optical emissions*: Optical ultraviolet (UV) signals are produced as a result of various ionization, excitation and recombination processes during PDs. Every material emits light of different wavelengths as a result of these phenomena.
(4) *High frequency current components*: PD is basically a surge of electrons and hence a current pulse. These current pulses are superimposed on the normal load current. These pulses have very small rise time and high frequency.
(5) *Thermal emissions (infrared emission and thermal ionization)*: The heating phenomena are mainly caused by increased resistance of loose contacts, series arcing across bad contacts or terminations. Ionization, excitation of atoms, and recombination of ions due to PDs also cause thermal emissions. Heat produced due to these phenomena lie in the range of infrared spectrum of electromagnetic radiation.
(6) *Chemical emissions*: During the process of ionization a neutral item or molecule loses or gains electrons thereby acquiring a net charge. These ions combine with the ions of other atoms to produce a molecule which is called by-product of PD. Most common by-products of PD's in the air insulated switchgears are ozone and nitrogen oxide. When they react with the moisture and water molecules, they form nitric acid [19]. Nitric acid is corrosive in nature and may cause severe damage to the contacts as well as dielectrics and insulators. It plays a major role in the decomposition of the chemical structure of the insulating material.

4.5. Sensor technologies

Many sensor technologies can be applied for detecting the above mentioned pre-arc phenomena. A short summary and evaluation of the technologies is presented in the following.

(1) *RF antenna*: Antenna type sensors are widely used for PD detection in the higher frequency range (HF/VHF/UHF).
(2) *Coupling capacitor*: Coupling capacitors are used to transfer partial discharge energy from electrical network to the measurement setup. Sometimes they are used as proximity sensors for current or voltage measurement. The main disadvantage is that the capacitors have to be designed in order to withstand 50/60 Hz rated voltage level of the equipment, and should be

manufactured to have low inductance in order to have good high-frequency response [20,21].

(3) *High frequency current transformer (HFCT)*: The working principle of HFCT is the same as normal 50/60 Hz current transformer. Partial discharges induce a current proportional to the current of the discharge. HFCT sensor is one of the most popular inductive sensors in condition monitoring technologies for most kind of applications in power systems due to its portability, cost effectiveness, and non-intruding characteristics [22].
(4) *Rogowski coil*: The air cored sensor is placed around the conductor, where current pulses caused by partial discharge are to be measured. The changing current induces voltage in the coil. Rogowski coils can be designed for the desired frequency. They are immune to external interferences [23].
(5) *Piezoelectric ultrasonic sensor*: Piezoelectric sensors are directional sensors that can detect PD in air insulated switchgear. Internal discharges in insulation do produce ultrasound signals, but the signal may get attenuated through the insulation and through air. Thus piezoelectric sensors are not optimal for internal PD detection due to lower sensitivity and directionality [24,25].
(6) *Ultraviolet sensor*: Ultraviolet sensors are sensitive to ultraviolet light, emitted by PDs in insulation material or arcing across loose contacts. Mal-operation is possible due to the availability of other light sources, if this technique is implemented alone.
(7) *Thermal sensors*: There is a wide variety of thermal sensors. Some sensors work on the principle of resistance change of the material, whereas others work on electron emission due to heat. Such sensors are less sensitive. If these sensors are kept in contact with the insulation, they may melt themselves due to excessive heat produced by arcing. If they are kept at a larger distance, they may not sense the temperature of the hot spot accurately, but rather sense the temperature of the surrounding air only [26].

There are some sensors which can measure infrared radiations emitted due to heat. Conventionally, infrared thermal sensors and thermographs were widely used to locate the hot spot created due to loose contacts and partial discharges. Such IR thermal cameras are very expensive and not practical to implement for online condition monitoring. In the market there are infrared sensors which are very small in size, accurate, and easy to install in the switchgear permanently for the online monitoring purpose [21]. Some manufacturers have implemented thermal ionization detectors which detect the presence of certain ions produced by thermal effects in the switchgear [26].

Table 1 presents a summary of the phenomena, sensor types, and applicable switchgear type.

Table 1
Summary of sensor technologies.

Phenomenon	Sensors	Switchgear type
Electromagnetic emissions	RF antenna	MV
Acoustic (ultrasonic) emissions	Piezoelectric probe	MV
Ultraviolet emissions	Optical sensors	MV
High frequency PD pulses	HFCT, RC, CC	MV
Thermal emissions	Thermocouples, IR sensors, TID	MV, LV
Chemical emissions	Analyzers	MV, LV
Fast variation in electric field	Differential electric field sensor	MV

5. Reactive protection

5.1. Why reactive protection is needed

Preventive or predictive measures are not able to eliminate the risk of all arc faults. Causes like direct human interaction or equipment malfunction may still lead to serious faults. This is the reason why reactive protection, i.e. reacting to phenomena detected after arc ignition, is always justified. However, traditional overcurrent protection is fairly inefficient in arc-flash faults because it is too slow and leads to long arcing time and high released energy.

5.2. Arc-flash incident energy

Incident energy is a concept defined in [27]. Its primary purpose is to provide means for evaluating needed safe working distances and personal protective equipment (PPE) for employees. However, incident energy calculations can also be used when estimating potential damage to equipment, and especially when different protection approaches are compared.

Incident energy of an arc-flash incident depends on voltage, distance, current and arcing time [27]. In practice, the key factors are arcing current and arcing time. Some protection approaches emphasize limitation of current while most solutions aim at short arcing time.

5.3. Arcing current

Fault arc is a highly random phenomenon. During arcing the length of the arc varies and the arc resistance is not constant. However, this is not the only uncertainty related to the estimation of arcing current. The magnitude of arcing current is often derived from bolted fault current, but the bolted fault current is not a constant either. Both the maximum and minimum values are needed for arc-flash studies [27]. The highest incident energy can be caused by the highest or lowest bolted and arcing fault current. This is because lower arcing fault current often leads to longer trip time [27–29].

According to [30] the initial symmetrical short-circuit current is calculated:

$$I_k'' = \frac{cU_n}{\sqrt{3}Z_k} \quad (1)$$

where c = voltage factor (0.95, 1.00, 1.05 or 1.10); U_n = nominal system voltage (line-to-line); Z_k = short-circuit impedance

Fig. 4 presents the system diagram and the equivalent circuit diagram related to Eq. (1). It illustrates the importance of the impedance of the feeding network (consisting of R_{Qt} and $\underline{X}_{Qt}$).

The impact of available fault current variations on arc-flash calculations has been studied in [28,31,32]. The available fault current data from the utility is not always available. In calculations the utility network is assumed to be infinite bus, but this leads to incorrect estimation of minimum fault current. The source impedance can vary because of changes of network connection or connected generators. Often the connection of renewable energy sources leads to lower short-circuit power. While conventional synchronous generators are able to provide 5–10 times their nominal current, photovoltaic systems, based on power electronics, provide fault currents only close to their nominal current [28]. For example, in Denmark the high share of wind power has led to reduction of short-circuit power [33].

The available short-circuit power from the utility has major influence on short circuit current on MV level but minor impact on LV (<1000 V) level because of the high impedance of MV/LV transformers. The impedances of the MV/LV transformer and the LV conductors are very dominating in the circuit. Often infinite source

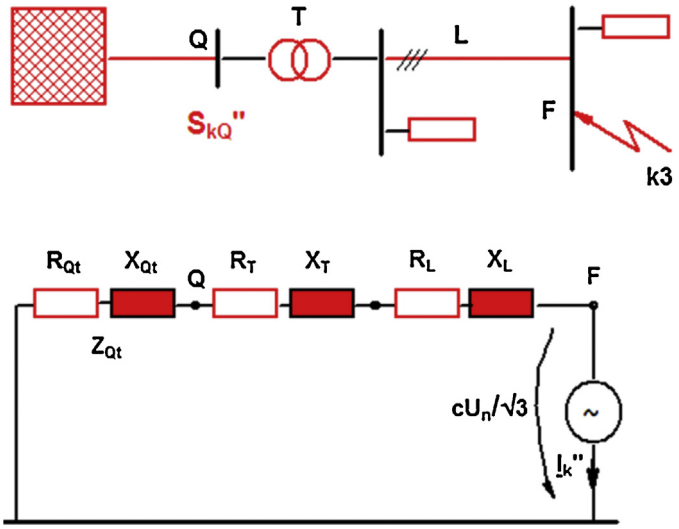


Fig. 4. System diagram and equivalent circuit diagram for calculating short-circuit current according to [26].

impedance on the transformer primary has been applied in short-circuit current calculations, but this does not guarantee the most conservative incident energy calculations [31].

5.4. Current-limiting approach in arc-flash protection

Limitation of arcing current leads to reduction of incident energy, provided that the arcing time is not significantly affected. However, in spite of being widely reported, the fact that lower current can cause higher incident energy is probably not adequately understood.

The fault arc always has some impedance, and the arcing current is lower than in a fault without fault impedance (bolted fault). For high-voltage arcs, the impedance of the arc is very low, and the arcing current is practically the same as the bolted fault current, and it can be calculated according to e.g. IEC standard 60909 [34]. Especially for applications under 1000 V, the impedance of the arc is often significant, and the arc fault current is lower than the bolted fault current [27]. According to [35], the arc fault current can typically fall between 20% and 40% of the prospective or bolted fault current, and [34] confirms that the arc fault current can be as low as 33% of the calculated bolted fault current at any particular location per [36].

The fact that arcing current can be significantly lower than the prospected current sets challenges to protection. The protection must be able to detect the fault, and what is especially important in arc fault incidents, is that it must be able to operate very fast.

The impact of the reduction of arcing current should not be analyzed without examining the impact on operation time of the protection, the arcing time and the released energy. When inverse time overcurrent protection is applied the faults with high fault currents will be cleared much quicker than the low current faults. According to [32], the largest arc energy with the normal time graded protection is normally produced by the minimum arcing fault current. In the tests reported in [29] 361 out of 869 locations had incident energy calculations determined at the low range of the estimated current. The behavior of current-limiting (CL) fuses depends on the current level. When the current is in the current-limiting range, CL fuses are very effective in reducing the incident energy level, the pressure wave and even the mechanical stress caused by the high fault current. When high current can be taken for granted, CL fuses are an excellent solution. However, current limiting fuses provide current-limiting action only in case of very high fault currents [37,38]. When used in combination with switching devices the maximum arc energy may occur at current levels below the maximum interrupting rating [1]. The risk of higher

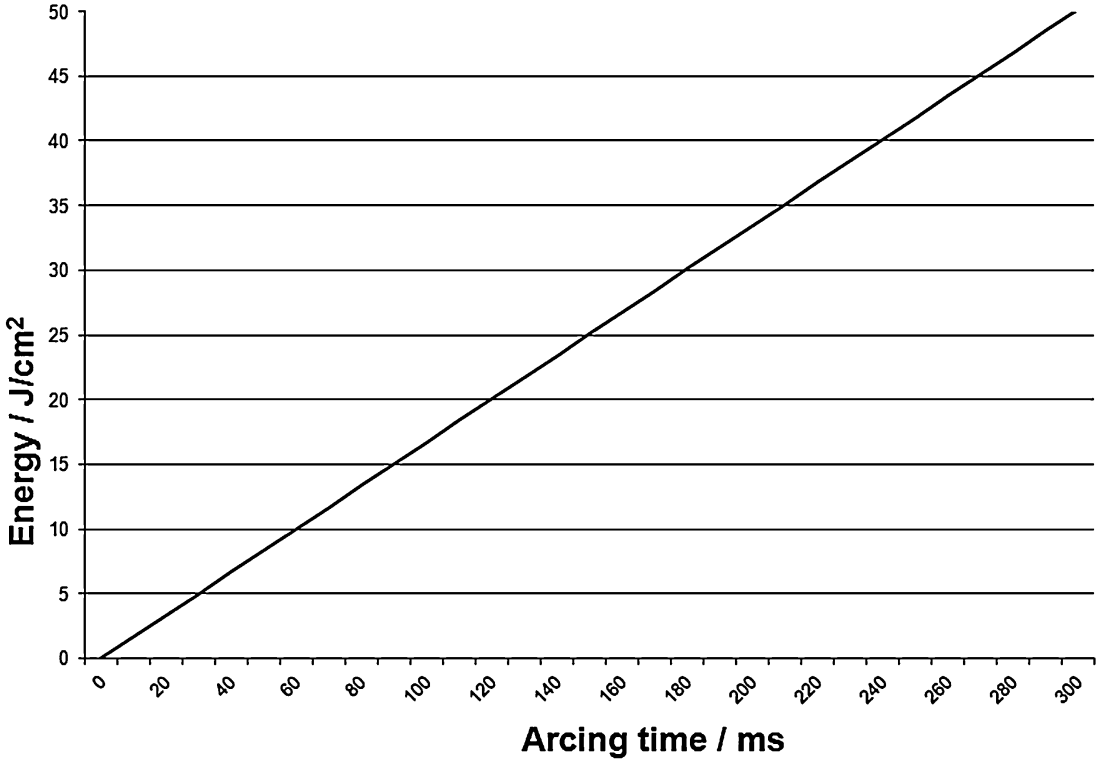


Fig. 5. Incident energy as a function of time (according to [24]; U = 15 kV, I_k = 50 kA, gap 152 mm, working distance 910 mm).

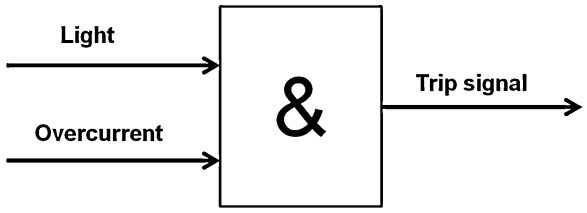


Fig. 6. The basic principle of light and overcurrent based arc-flash protection.

incident energy level because of lower current is also clearly illustrated in the test result figures of [27] regarding CL fuses.

Differences in arcing current magnitude can also be caused by various electrode configurations. In tests reported in [29], significantly lower arc currents than predicted by the equations in [27] were measured. Because of higher clearing times of overcurrent protection, incident energy calculations can be greater than three times the standard calculations [29].

Another significant risk caused by low fault current is that the protection may not properly respond or even trip at all [37,39]. Calculation of minimum fault current is essential.

In conclusion, current-limiting approach to mitigate the impact of arc faults includes many uncertainties and risks related to the arcing time and incident energy.

5.5. Limitation of incident energy by reducing arcing time

Incident energy is directly proportional to the arcing time. This is illustrated in Fig. 5. Reduction of arcing time has become the dominant approach in arc-flash mitigation. Various technologies have been presented in [40].

Compared to traditional overcurrent protection, protection time in busbar faults can be improved by zone-selective interlocking. Traditional busbar differential protection provides better performance with approximate 1 cycle operation time. However, busbar differential is not effective if the fault occurs in cable compartment, typically in cable terminations. In some countries busbar differential protection is considered as obsolete technology for arc-flash protection.

A maintenance switch, activating instantaneous tripping when maintenance work is carried out, provides very fast tripping, but it is primarily intended for personnel protection only.

5.6. Light and overcurrent based protection

The rising technology to reduce arcing time is based on simultaneous detection of light and overcurrent. Together, these two conditions provide an extremely fast and very secure arc flash detection scheme [41]. Along with detection of phase overcurrent, zero-sequence overcurrent detection can be applied for phase-to-ground faults. Fig. 6 illustrates the basic operation principle of light and overcurrent based protection.

The light of the fault arc is detected by sensors sensitive to light. Both point type of sensors and loop type of fiber sensors are commonly applied. Point sensors monitor a restricted area, e.g. cable termination compartment, and they are able to indicate the fault location rather accurately enabling selective protection. Fiber optic sensors typically monitor larger areas, such as the whole busbar area or several circuit breaker compartments [42].

The sensitivity level of the sensor is not a critical issue because the light intensity caused by an arc fault is normally very high compared to ambient light intensity, and because of the overcurrent condition. Practice has shown that the only situation where special attention has to be paid to the detection of light is the protection of the circuit breaker compartment of LV systems, if the CB emits

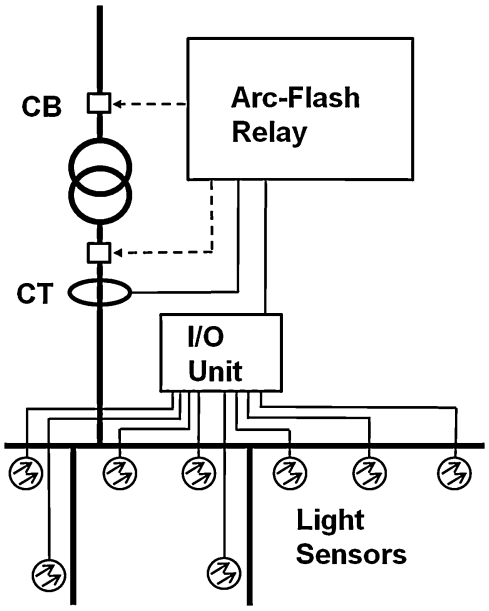


Fig. 7. An example of a light and overcurrent detection based arc-flash protection system.

light. Fig. 7 presents a scheme of a simple arc-flash protection system including current measurement, light sensors and the I/O unit for sensors, and the arc-flash relay.

One of the advantages of light detection is the lack of a requirement to coordinate with downstream devices and the ability to operate extremely fast (½ cycle or less) [41]. Another benefit is that retrofit installation is fairly easy when point type of sensors are applied.

Existing current transformers can be utilized in detection of overcurrent. Like the light from the fault arc, the overcurrent can be detected within 1 ms. Thus, the arc detection time using light and current based detection is approximately 1 ms.

Light and overcurrent based protection can be integrated into common numerical protection relays or it can be applied by using dedicated arc-flash protection relays. State-of-the art arc-flash relays provide very short trip time, several protection zones for selective tripping, and versatile communication options. When applying this technology, almost all the arcing time consists of the operation time of the primary device responsible for arc elimination, i.e. normally the circuit breaker or in some cases the arc eliminator. Arc eliminator is explained in Section 6.

Fig. 8 presents a comparison of incident energy levels of a MV system case, illustrating the impact of arcing time. Arcing time 67 ms of the example corresponds in practice to a system with light and overcurrent based protection tripping circuit breaker, and the 5 ms case illustrates the efficiency of protection system enhancement by an arc eliminator.

When applying current and light base arc-flash protection, selective protection can be achieved, especially with optical point sensors indicating the fault location. This means that e.g. in cable termination faults, only the breaker of the outgoing feeder will trip while the rest of the systems remains energized.

6. Fault arc elimination

6.1. Elimination technologies

There are three basic technologies for arc elimination: fuses, circuit breakers, and arc eliminators. CL fuses have been discussed

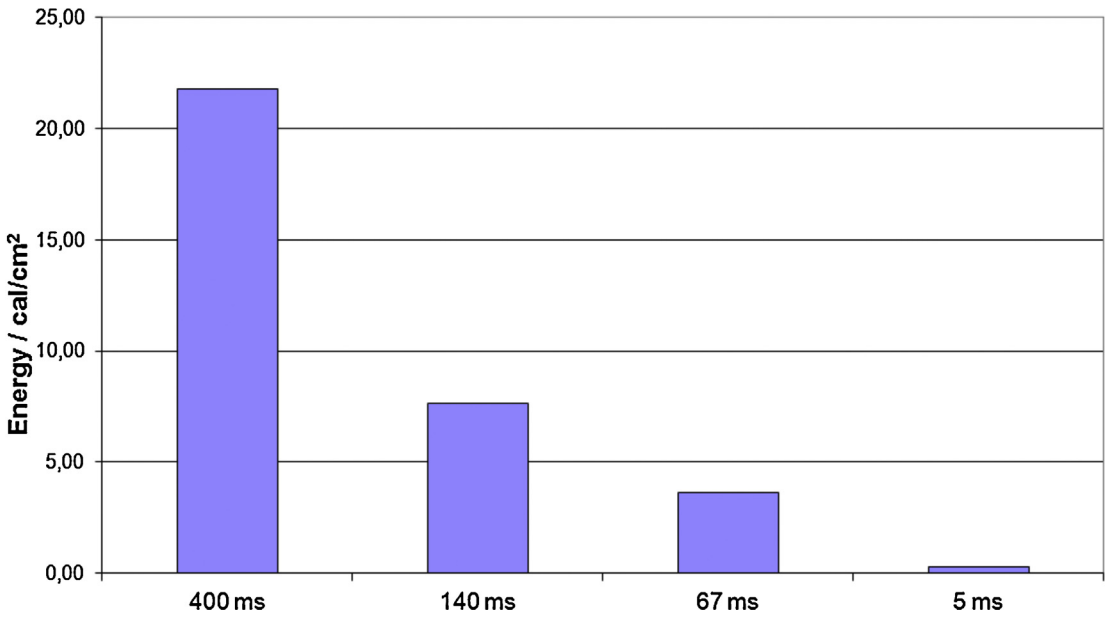


Fig. 8. Comparison of incident energy levels with different arcing times (according to [24]; U = 15 kV, I_k = 40 kA, gap = 152 mm, distance = 910 mm).

above. The focus of the following subsections is on CB and arc eliminator technology.

6.2. Circuit breakers

For light and overcurrent based protection, the relay trip time is typically less than half cycle. When semiconductor output (IGBT, Insulated Gate Bipolar Transistor output) is used instead of mechanical output relay, trip time can be as short as 1 ms. Since operation times of circuit breakers are some tens of milliseconds, it is clear that the arcing time depends practically on CB operation time only.

Good questions have been raised concerning CB's:

- What is the actual operation time, i.e. should a few cycles be added to the value given by the manufacturer?
- What if the CB fails?

According to [35], CB operation time is the maximum time, and many breakers actually trip faster than what is in the datasheets. This is perfectly in line with experience gained in switchgear arc fault testing. Conservative value for total arcing time is thus the sum of relay time and CB time, the latter being the dominant part. This aspect should be taken into consideration when selecting CB's.

CB is the component that finally breaks the current. Thus it is most important to ensure that the CB's are in working order. Since no device can be 100% reliable, circuit breaker failure protection (CBFP), tripping the upstream breaker, should be applied.

6.3. Arc eliminators

An arc eliminator is a device which eliminates an arc extremely fast by creating an intentional parallel short circuit. The technology is also called crowbar technology, arc quenching or high speed grounding. So far arc eliminators have not been widely applied, although the technology is listed as an option to provide highest possible level of protection in IEC standard [1]. One of the most interesting application areas is retrofit installation in switchgear which is not arc resistant. A very significant improvement of both personnel safety and mitigation of equipment damage can be achieved. In addition to minimization of incident energy, arc eliminator also mitigates the pressure impact. The effectiveness of arc eliminator technology against the pressure wave has been illustrated in [43,44].

The operation principle is simple: when an arc fault is detected, the device will create an intentional high speed short circuit in the system so that the voltage collapses and the arc is extinguished. Various manufacturers have different technologies in the methods related to the creation of the short circuit, e.g. pyrotechnical pressure element, Thomson coil, micro gas cartridges, or a spring mechanism assisted by an electromagnetic repulsion system.

Arc eliminators are used in combination with light and overcurrent based arc-flash detection. The arc detection system is able to trip the eliminator within 1–2 ms, and the high speed primary circuit device operates within a few milliseconds. Fast communication between the arc-flash protection relay and the short-circuit device is necessary. A typical arcing time is less than 5 ms. The arc-flash relay also sends a tripping command to the circuit breaker, and the CB eliminates the short-circuit current within a few cycles. Since the arcing time is minimal, hazard and damage caused by arc fault are almost nonexistent. Of course, a power system interruption is caused when an arc eliminator operates.

Short-circuit devices have been opposed because of the concern on the high current they cause. Of course high current can be detrimental to transformers, but the concern seems to be exaggerated. In MV systems the arcing current is almost equal to bolted fault current. Thus, the creation of an intentional, symmetrical short-circuit does not significantly increase the current. In LV systems, there are options to reduce the level of the current. A combination of an arc quencher and CL fuses can be used, or a small but current tolerant resistor can be added to the circuit in order to lower the stress to power system components.

When estimating the risks caused by the high current, two additional issues should be kept in mind. Transformer failures caused by short-circuit events are relatively rare, and according to IEEE and IEC standards, transformers shall be designed to withstand the electromagnetic forces and the thermal stresses produced during the flow of a short-circuit current [45].

Additionally, since it is very difficult to find any evidence or experience of real cases where arc eliminators had caused damage to equipment, the risk level related to high current caused by arc eliminators can be estimated acceptable [46].

7. IEC 61850 based communication in arc-flash protection systems

In arc-flash protection systems, the speed and reliability requirements for communication are very high, because every millisecond increases the released energy and leads to larger damage. Exchange of information is essential not only between the relays but also between the sensors detecting the arc and the relays, and between the relays and the circuit breakers. One example of a successful approach is minimization of the transmitted information and the use of high-speed serial protocol. Even selective protection is possible, when the transmitted bit includes the information on the activated sensor and thus the protection zone [47]. Communication delays are very low, enabling less than 2 ms total detection and trip time.

Ethernet based communication, in particular IEC 61850 [48] based technology, is widely applied in protection systems but in arc-flash protection systems it is so far rather rarely deployed. However, when carefully applied, adequate performance and reliability can be achieved along with the benefits this technology provides.

Generic Object Oriented Substation Event (GOOSE) messages over Ethernet are a standardized method for communication in protection applications. Zone-selective interlocking (ZSI, reverse interlocking) is a common application related to arc-flash protection where IEC 61850 and GOOSE have successfully been utilized [49]. GOOSE messaging reduces coordination time and simplifies relay protection coordination [50]. However, ZSI is slower than light and overcurrent based arc-flash protection [51].

In [52], GOOSE messages are limited to relay-to-relay communications in light and overcurrent based arc-flash protection system. GOOSE messaging can also be applied in the communication between other components of the arc-flash protection system: sensors, input/output units, relays, and circuit breakers.

In order to increase reliability and to avoid delays caused by network traffic, virtual local area networks (VLAN) for separated traffic can be established [50,53]. Another means to enable very fast communication is to utilize high speed fiber media for connection to devices [47,54]. It has been shown that the speed of GOOSE based communication is as good as in direct serial communication [53].

There are several important benefits of applying IEC 61850 standard based GOOSE messaging for arc-flash protection. It allows standard Ethernet components and longer distance between the components when fiber optic interface is applied. The standard based solution also is easier to monitor than manufacturer specific solution, and it enables multi-vendor systems. One of the major benefits is simplified and reduced wiring which also reduces costs [51].

8. Conclusions

This paper has given a holistic view of arc-flash mitigation technologies discussing both passive and active methods. Special attention has been paid to proactive on-line monitoring techniques that enable detection of slowly developing faults. So far these methods have not been widely applied. Since these techniques are not effective in all faults, reactive protection is justified.

Incident energy levels can be reduced significantly by reactive protection techniques having extremely short operation times or by limiting the fault current. Time-limiting approach has been preferred to current-limiting approach because the latter includes many uncertainties and risks related to arcing time and incident energy. In the future, arc-flash protection can be maximized by a combination of passive methods, proactive and reactive techniques, and enhanced by IEC 61850 based communication.

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